2. Features of SiC SBD

2.1 Device structure and features

With SiC, high breakdown voltage diodes above 1,200 V can be realized using the Schottky barrier diode (SBD) structure (up to approximately 200 V with Si-based SBD).

As a result, as shown in Figure 2-1, the recovery loss can be significantly reduced by replacing currently prevailing high speed PN junction diodes (FRD: fast recovery diode) with SBD. This contributes to improvement in the power supply efficiency, downsizing of passive components including coils by high frequency drive, and noise reduction. Their applications are mainly power factor correction (PFC) circuits and secondary side rectifying bridges, and extending to onboard chargers for EV, power conditioners for solar power generation, power supplies for servers, air conditioners, and so on.

Currently, SBD with breakdown voltages of 650 V, 1,200 V, and 1,700 V are listed in ROHM's lineup.

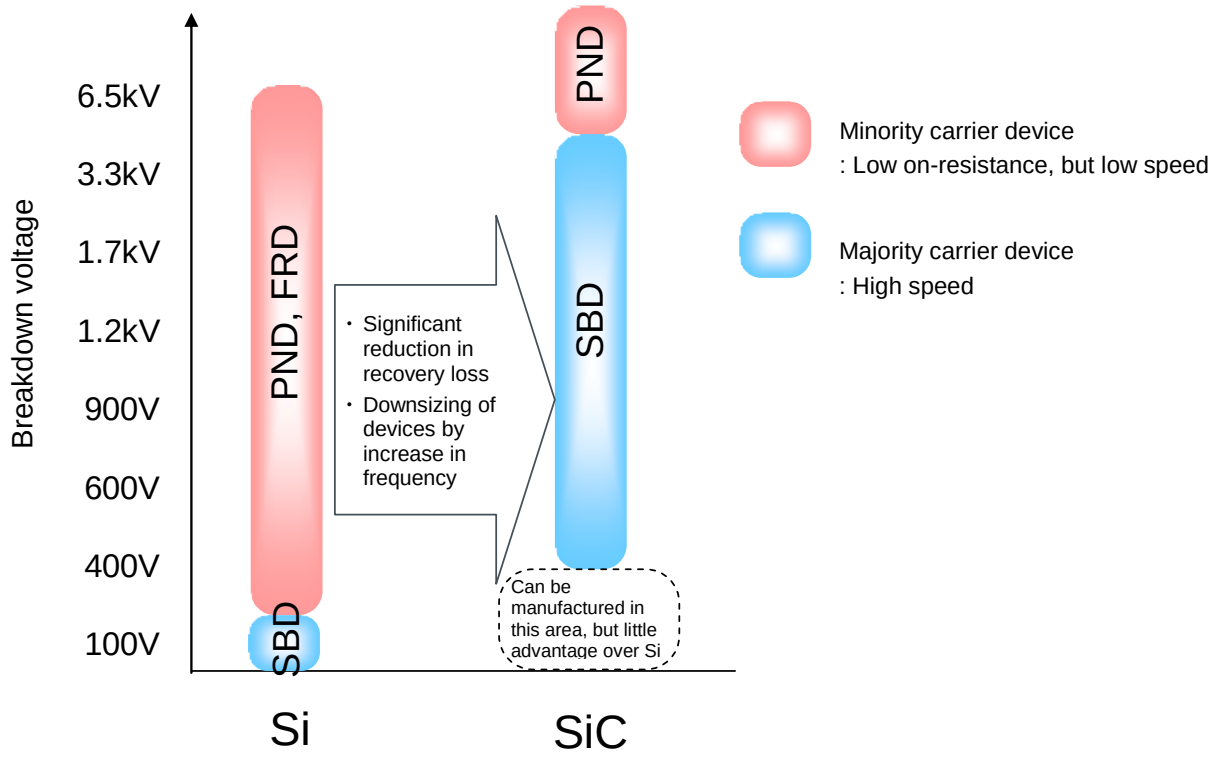


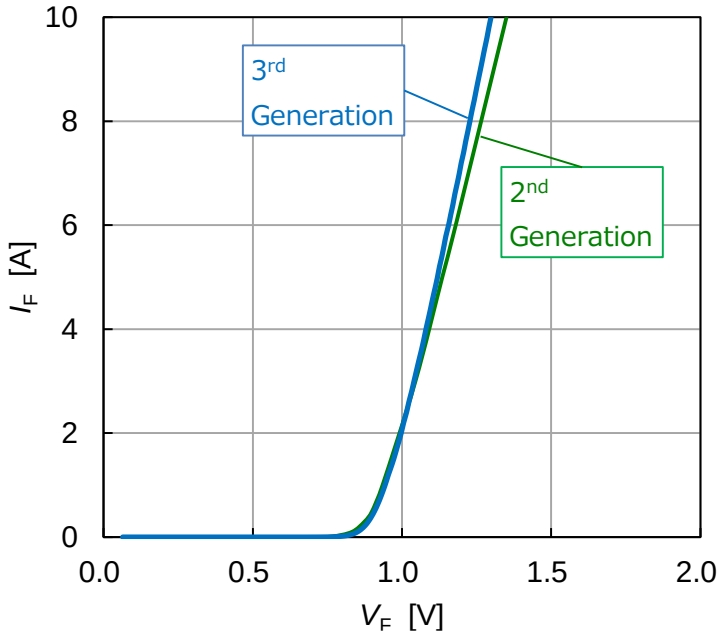
Figure 2-1. Rated voltage ranges for Si and SiC (diodes)

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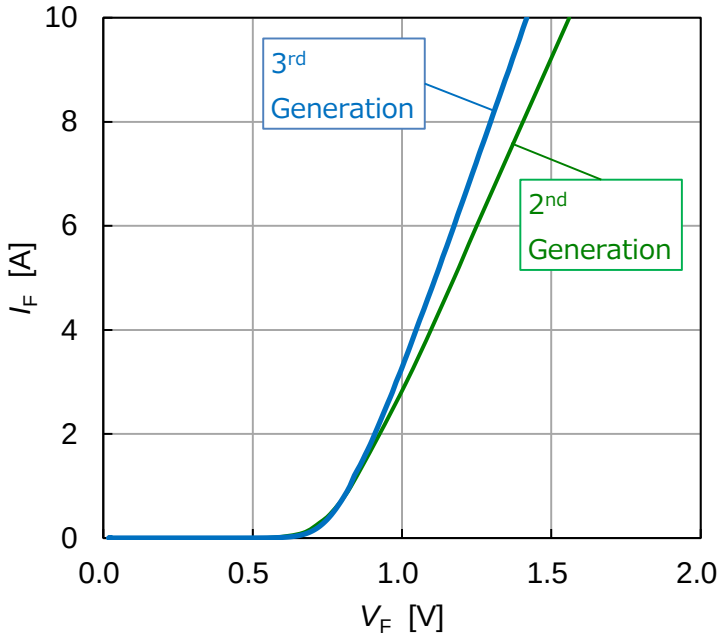
2.2 Forward characteristics of SiC SBD

The rising voltage of SiC SBD is a little less than 1 V, similar to Si FRD. The rising voltage is determined by the height of the Schottky barrier and can usually be reduced by designing a lower barrier height. However, there is a trade-off relation between the rising voltage and the leakage current in a reverse bias condition, which increases as the barrier height decreases. In the second generation of ROHM's SBD, innovations in processes successfully reduced the rising voltage by approximately 0.15 V while maintaining the leakage current and the recovery performance at the same level as conventional products. Furthermore, in the third generation SBD, V_F and the leakage current are further reduced by combining the JBS (junction barrier Schottky) structure and the low V_F process for the second generation SBD. In particular, V_F is significantly reduced at high temperature.

The temperature dependence differs from that of Si FRD, showing increase in V_F due to increase in the operating resistance as temperature increases. They can be safely used in parallel connections because a thermal runaway is unlikely to occur. Figure 2-2 shows typical V_F-I_F characteristics.



(a) Tj=25℃



(b) Tj=150℃

Figure 2-2. Forward characteristics of SiC SBD (650 V, 10 A class)

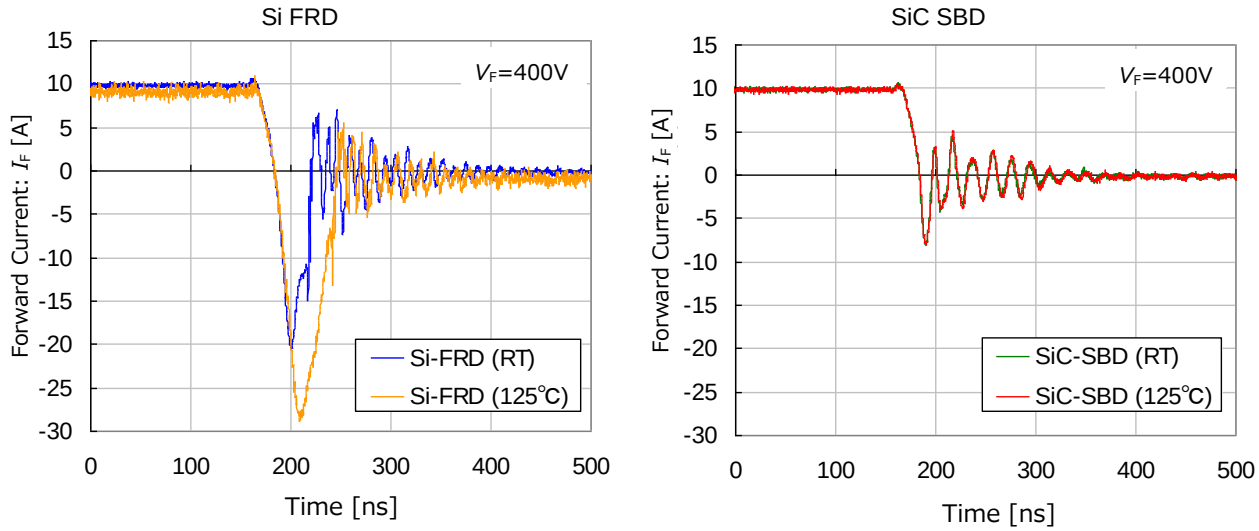
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2.3 Recovery characteristics of SiC SBD

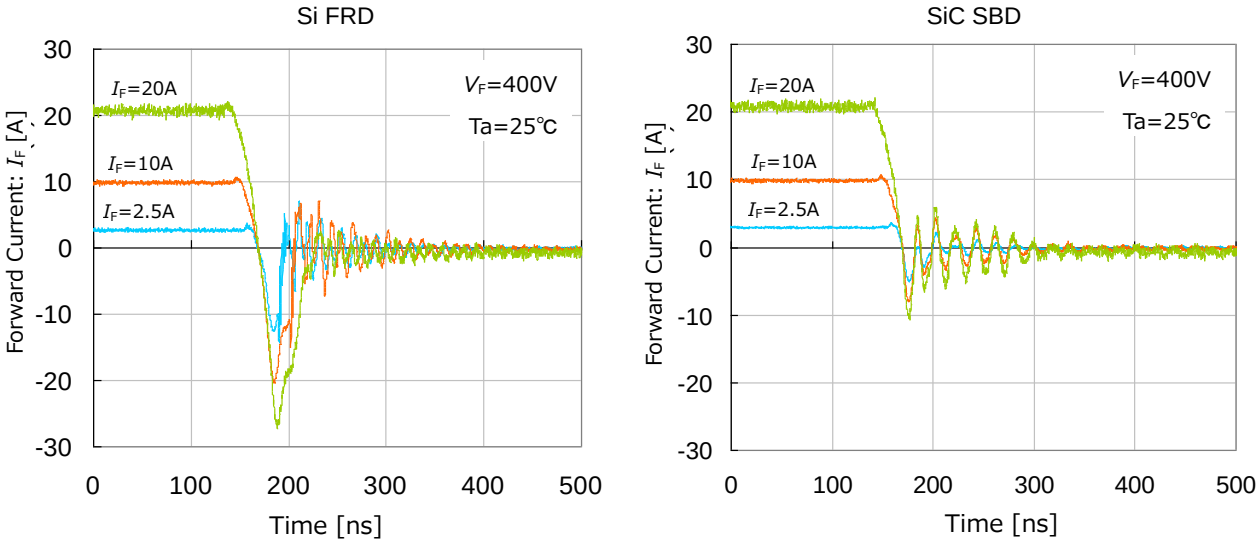
In Si-based high speed PN diodes (FRD: fast recovery diode), a large transient current flows momentarily when the direction switches from forward to backward, causing a large loss due to transition to the reverse bias state during this period. This is attributed to a contribution to electrical conduction by minority carriers that have been stored within the drift layer during a forward conduction until they disappear (storage time). The larger the forward current and the higher the temperature, the longer the recovery time and the larger the recovery current, resulting in a significant loss.

In contrast, since SiC SBD are majority carrier devices (unipolar devices), which do not use minority carriers for electrical conduction, no accumulation of minority carriers occurs in principle. Only a small current flows that is sufficient to discharge the junction capacitance, and the loss can be significantly reduced compared with Si FRD. Since this transient current is mostly independent of the temperature and the forward current, a stable and high speed recovery can be achieved in any environment. In addition, noise generated due to the recovery current is expected to be reduced.

Figure 2-3 shows the measurement results of the recovery characteristics of SiC SBD and Si FRD. It can be seen that the recovery current is significantly reduced with SiC, irrespective of the operating temperature and the flowing current.



(a) Temperature dependence



(b) Forward current dependence

Figure 2-3. Comparison of recovery characteristics (650 V, 10 A class)

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3. Features of SiC MOSFET

3.1 Device structure and features

With Si, the on-resistance per unit area increases with the breakdown voltage of a device (the on-resistance increases with the 2nd to 2.5th power of the breakdown voltage approximately). Therefore, IGBT (insulated gate bipolar transistors) are mainly used for a voltage of 600 V and higher. Compared with MOSFET, the on-resistance of the IGBT is reduced with conductivity modulation, in which holes (minority carriers) are injected into the drift layer. However, accumulation of the minority carriers generates a tail current at turn-off, resulting in a significant switching loss.

Since the resistance of the drift layer is lower for SiC compared with the Si devices, it is unnecessary to utilize the conductivity modulation. Therefore, a high breakdown voltage and low resistance can be simultaneously achieved in the MOSFET, which has a high speed device structure. In principle, the tail current will not be generated in the MOSFET. Therefore, replacing the IGBT with the MOSFET can significantly reduce the switching loss and downsize cooling structures such as a heat sink. The high frequency drive by the MOSFET can also contribute to downsizing of passive components, which is impossible with the IGBT. Furthermore, the SiC MOSFET also have advantages over the Si MOSFET in the range between 600 V and 900 V, such as smaller chip area (enabling mounting on smaller packages) and a very small recovery loss of the body diode.

Currently, the planar and trench type MOSFET with breakdown voltages of 650 V, 1,200 V, and 1,700 V are listed in ROHM's lineup. Their applications extend to various uses, including onboard chargers, power supplies for industrial equipment, and inverter and converter sections of high efficiency power conditioners.

Figure 3-1 shows a comparison between Si and SiC for the rated voltage ranges of the MOSFET and IGBT.

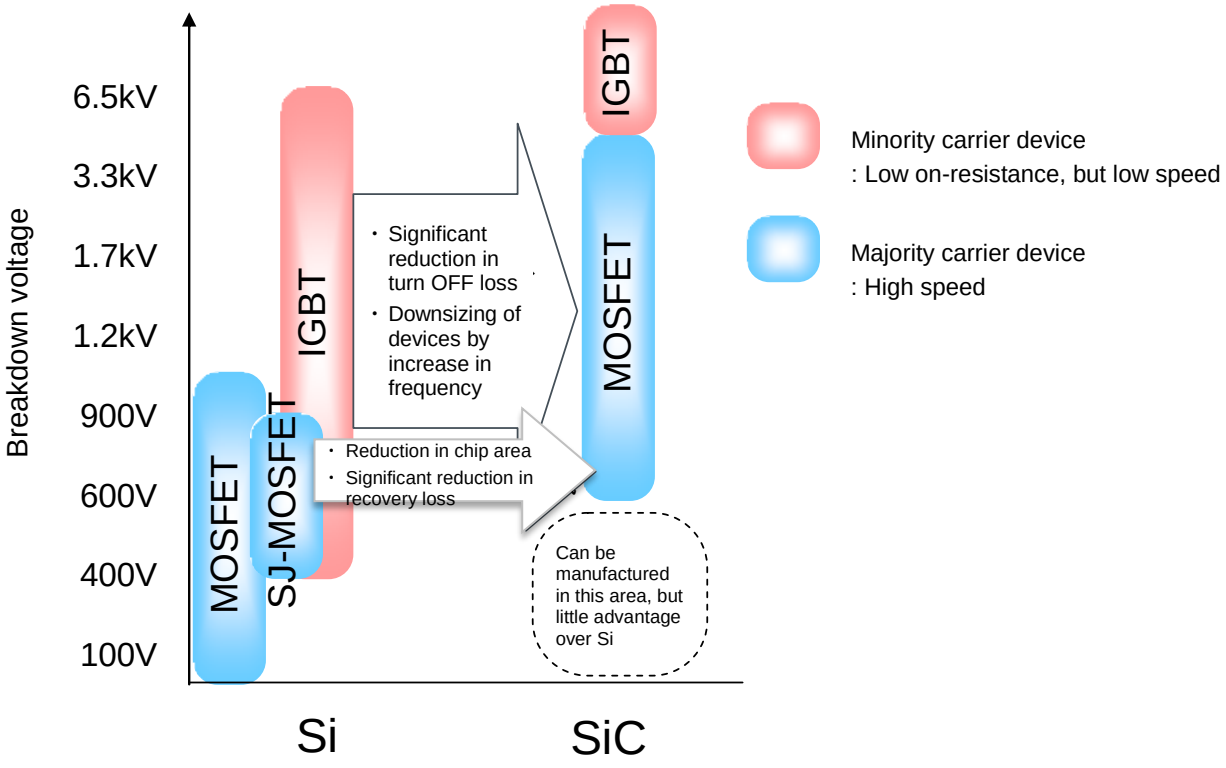


Figure 3-1. Rated voltage range for Si and SiC (MOSFET and IGBT)

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3.2 Standardized on-resistance (RonA)

Since the dielectric breakdown field strength of SiC is approximately 10 times higher than that of Si, high breakdown voltage can be achieved with a drift layer having a low specific resistance and a thin film thickness. Therefore, compared at the same breakdown voltage, a device with a smaller standardized on-resistance (RonA: on-resistance per unit area) can be manufactured. As shown in Figure 3-2, when compared at the breakdown voltage of 900 V, for example, the SiC MOSFET can realize the same on-resistance with a chip size approximately 1/100 and 1/10 of that of the Si MOSFET and the super junction (SJ) MOSFET, respectively. This enables a reduction of the on-resistance with a smaller package as well as the gate charge Qg and the capacitance.

For the super junction MOSFET, the breakdown voltage of existing products is only up to 900 V currently. For the SiC MOSFET, however, a breakdown voltage over 1,700 V can be achieved with a low on-resistance. Since the bipolar device structure (having a low on-resistance, but a slow switching) like the IGBT is unnecessary, a device that has a low on-resistance, high breakdown voltage, and high speed switching simultaneously can be realized with the SiC MOSFET.

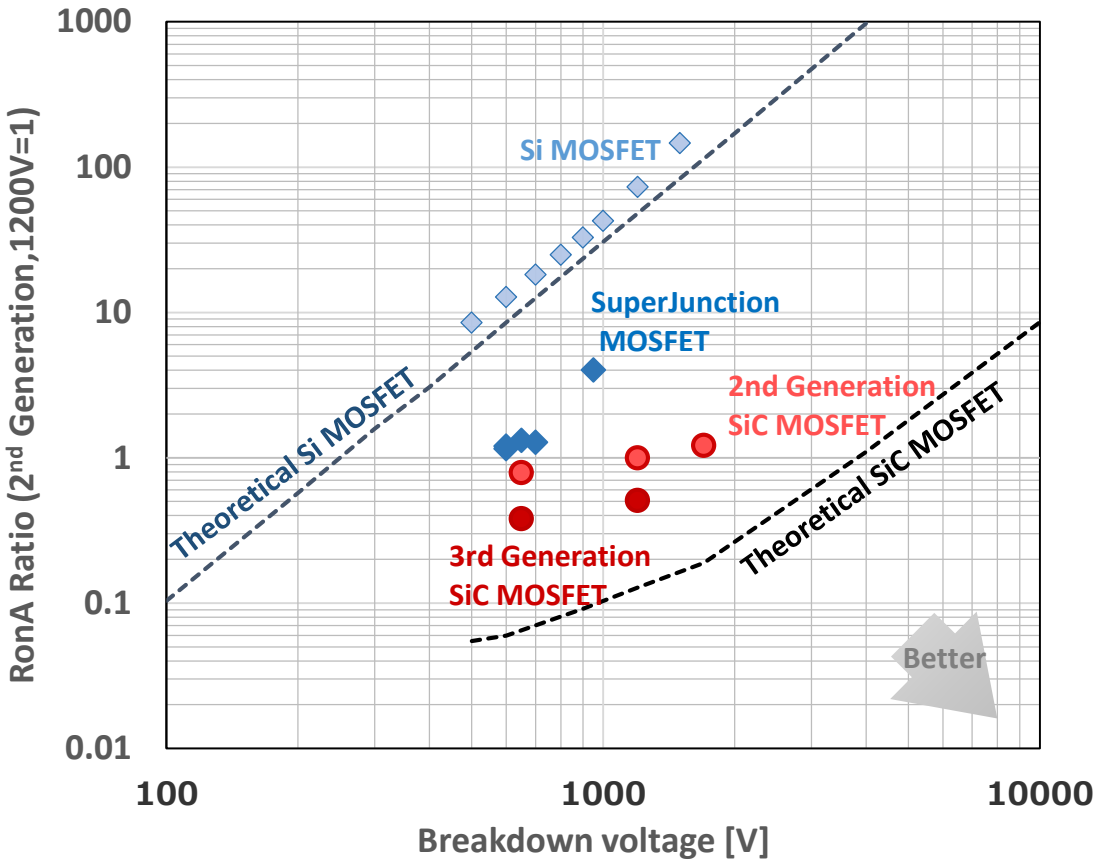
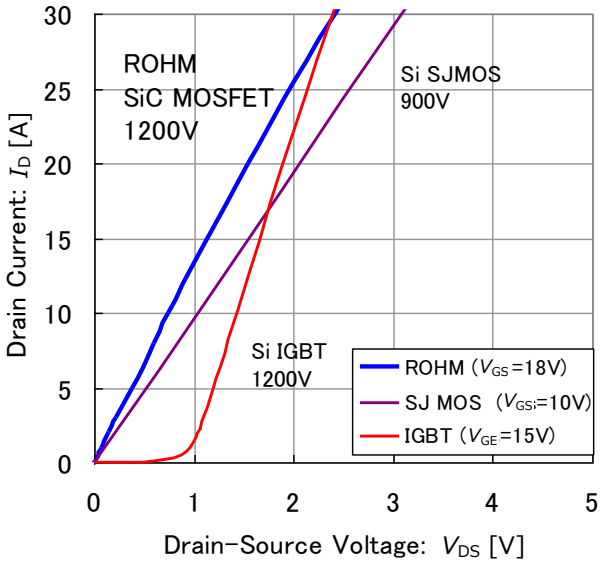


Figure 3-2. Comparison of RonA

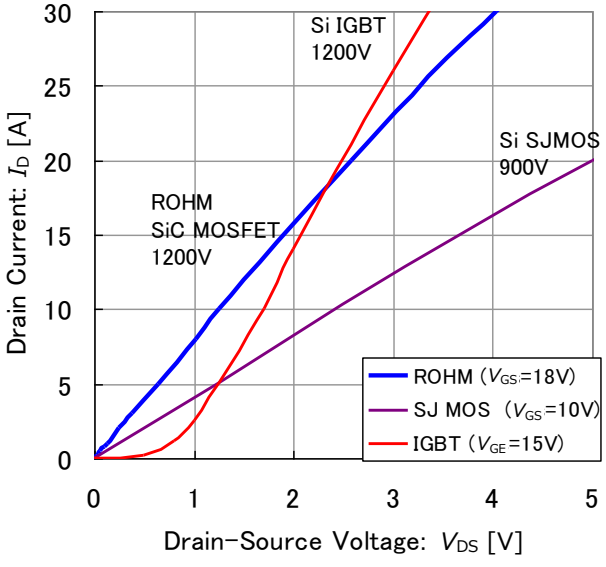
3.3 V_{DS}-I_D characteristics

Since there is no rising voltage like the IGBT, the SiC MOSFET can realize a low conduction loss in a wide current region from small to large current.

Furthermore, although the on-resistance of the Si MOSFET is increased by two times or more at 150°C from room temperature, a relatively low increase rate makes thermal design easier with the SiC MOSFET, and a low on-resistance can be realized even at high temperature. Figure 3-3 shows the V_{DS}-I_D characteristics of each device at ordinary and high temperatures.



(a) Ta=25℃



(b) Ta=150℃

Figure 3-3. V_{DS}-I_D characteristics

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3.7 Turn ON characteristics

SCH2080KE, where the SiC-MOSFET and SiC-SBD are mounted on the same package, and a product where the Si-IGBT and Si-FRD of equivalent classes are mounted on the same package are used to configure each half bridge circuit. Then, the switching waveforms are compared using the inductive load double pulse test (DPT). Figure 3-9 shows the test circuit.

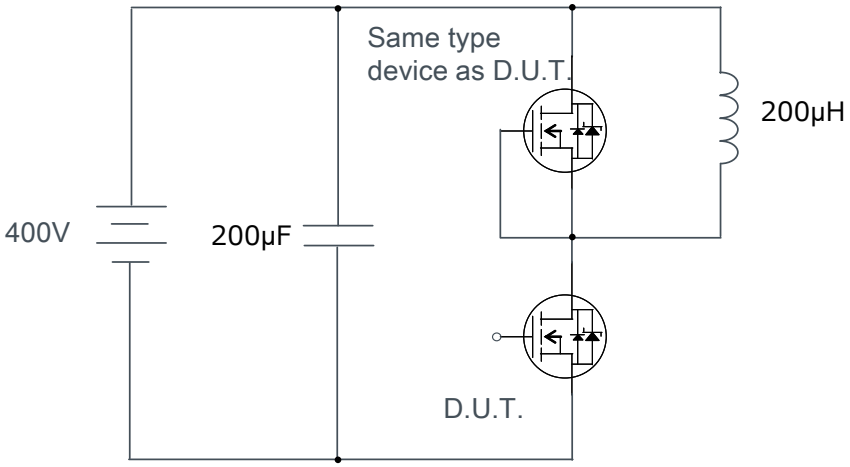


Figure 3-9. Double pulse test circuit

The turn ON speed of the SiC MOSFET is several tens of nanoseconds, similar to the Si IGBT and Si MOSFET. However, in the case of inductive load switching, a recovery current generated due to commutation to the diode of the upper arm penetrates and flows to the lower arm as well. Therefore, a significant loss is added depending on the performance of the diode. (Figure 3-10)

The body diode of the Si FRD or the Si MOSFET generally has a very large recovery current and generates a significant loss. In addition, this loss tends to increase further at high temperature. In contrast, high speed recovery is possible with the SiC SBD independently of temperature. The body diode of the SiC MOSFET also shows a high speed performance similar to the SiC SBD, although V_F is high. Such high speed recovery performance can reduce the turn ON loss (Eon) by several tens of percent.

The switching speed depends strongly on the external gate resistance R_{G_EXT}. To achieve high speed operation, a low gate resistance around a few ohms is recommended. Select an appropriate gate resistance with consideration for the surge voltage.

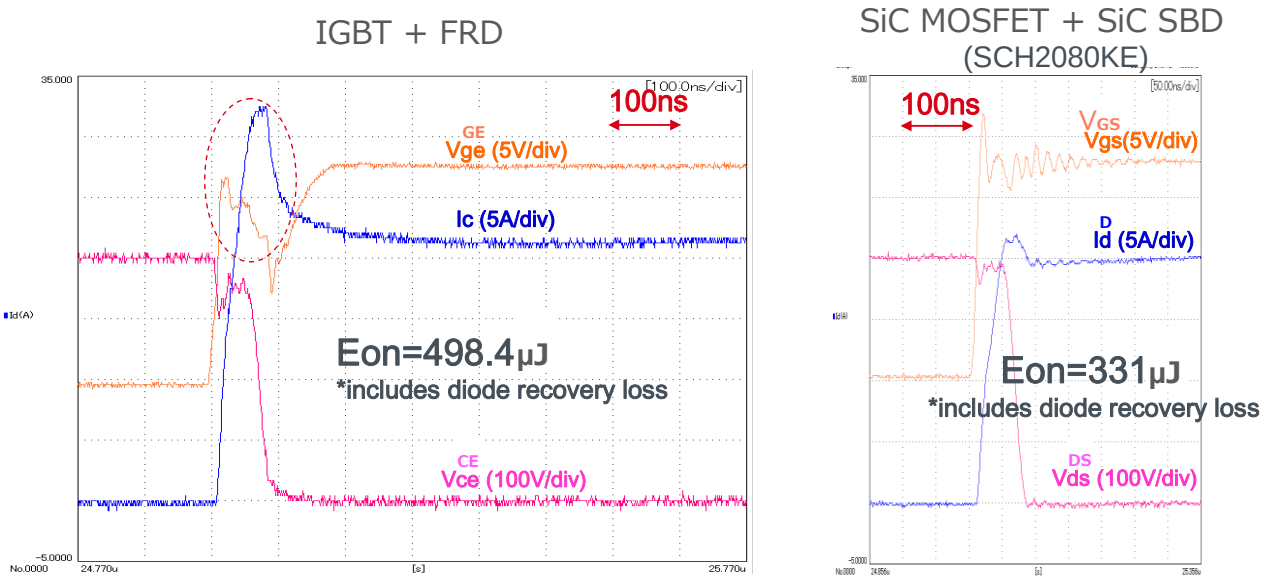


Figure 3-10. DPT turn ON waveforms

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3.8 Turn OFF characteristics

The most notable feature of the SiC MOSFET is that the tail current, which is observed in the IGBT, is not generated in principle. With SiC, the high speed MOSFET structure can be manufactured even at a breakdown voltage of 1,200 V or greater. Therefore, reduction in the turn OFF loss (Eoff) by approximately 90% compared with the IGBT (Figure 3-11) can be achieved, contributing to energy conservation of the circuit as well as simplification and downsizing of the cooling mechanism. While the tail current in the IGBT increases at higher temperature, almost no temperature dependence exists in the MOSFET.

Furthermore, the IGBT cannot usually be used at a high frequency region above 20 kHz because the junction temperature (Tj) exceeds the rating due to heat generated with a large switching loss. However, since Eoff is small, the SiC MOSFET can perform high speed switching operation at 50 kHz or higher. Operating at higher frequency enables downsizing of passive components including transformers and filters. (Figure 3-12)

The switching speed depends strongly on the external gate resistance R_{G_EXT}. To achieve high speed operation, a low gate resistance around a few ohms is recommended. Select an appropriate gate resistance with consideration for the surge voltage.

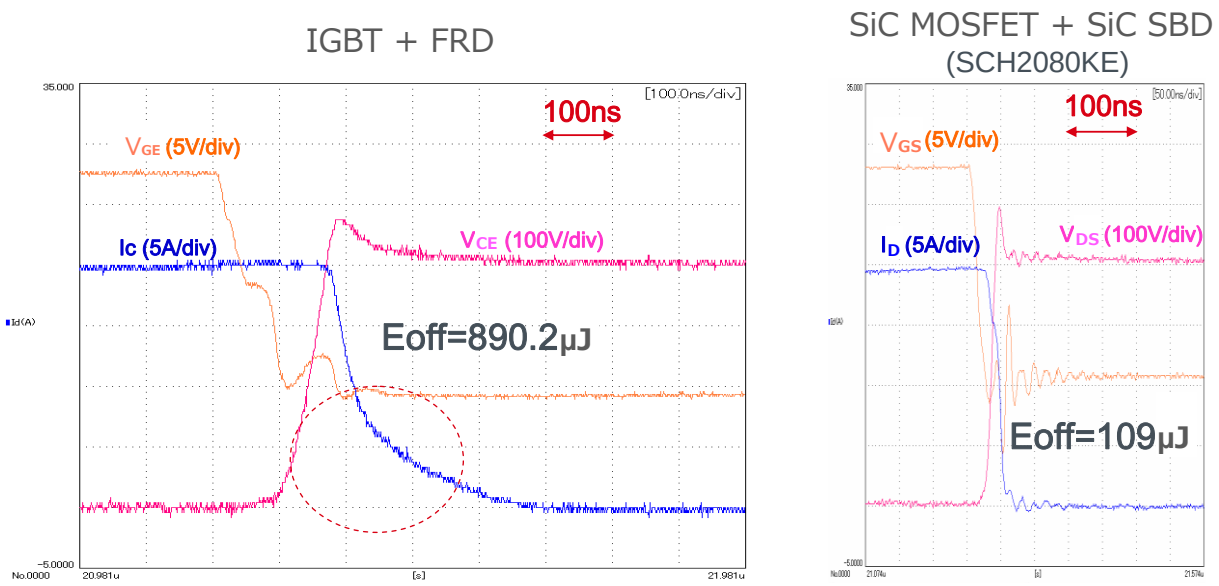
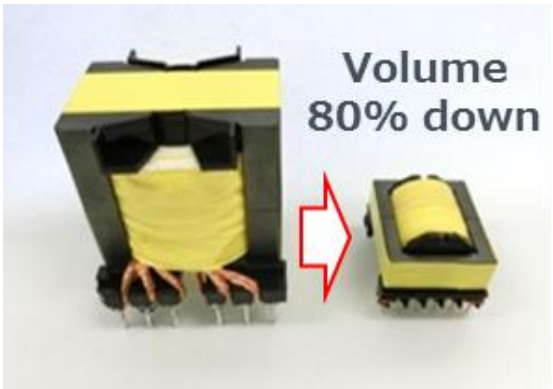


Figure 3-11. DPT turn OFF waveform



(a) 20 kHz (b) 160 kHz

Figure 3-12. Example of downsizing of transformer with high frequency operation

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